

Experiment Results

Safe Path Planning Under Perception Uncertainty

Scan Depths: 5 · 10 · 15 · 20 | Theta: 60° | Episodes: 500

1. Statistical Results

1.1 Correlation Analysis: Perception Entropy vs. Episode Reward

Spearman rank correlation (ρ) between Perception Entropy (H) and Episode Reward (R), Bonferroni correction applied across 8 tests.

Agent	Scan Depth	Spearman (ρ)	Original p	Adjusted p (Bonferroni)
Q-Learning	5	0.6074	5.56e-52	4.45e-51
Q-Learning	10	0.5134	1.73e-35	1.38e-34
Q-Learning	15	0.4907	1.99e-32	1.59e-31
Q-Learning	20	0.6322	3.62e-57	2.89e-56
SARSA	5	0.6139	7.85e-53	6.28e-52
SARSA	10	0.7077	3.59e-76	2.87e-75
SARSA	15	0.7042	5.52e-75	4.42e-74
SARSA	20	0.7539	6.45e-93	5.16e-92

1.2 Safety Margin Saturation — Mann-Whitney U Test

Non-parametric pairwise comparison ($\alpha = 0.05$) across consecutive scan depths.

Agent	Comparison	U-Statistic	p-value	Mean Diff	Result
Q-Learning	Scan 5 vs 10	120,238.00	0.297	+0.17	Not Significant
Q-Learning	Scan 10 vs 15	99,461.50	2.24e-08	+0.85	SIGNIFICANT
Q-Learning	Scan 15 vs 20	121,912.00	0.499	+0.23	Not Significant
SARSA	Scan 5 vs 10	86,407.50	2.89e-17	+1.33	SIGNIFICANT
SARSA	Scan 10 vs 15	128,815.50	0.403	-0.06	Not Significant
SARSA	Scan 15 vs 20	124,288.50	0.876	+0.04	Not Significant

1.3 Summary Findings

- **Correlation:** Highly significant positive correlation between perception entropy and reward across all settings ($p_{adj} < 0.001$).
- **Q-Learning saturation:** Safety performance saturates at Scan 15 (Scan 10 vs 15 significant; Scan 15 vs 20 not significant).
- **SARSA saturation:** Peak safety reached earlier at Scan 10 (Scan 5 vs 10 significant; subsequent comparisons not significant).

2. Key Figures
